Module 3: End of Module Assignment

TaskFlow - Task Management API ✅

Overview

Type: Mandatory Assignment
Estimated Time: 1-2 hours

Introduction

Build a simple TaskFlow API - a task management backend. You'll create a Spring Boot project and complete 3 tasks.

Project Setup

Step 1: Create Spring Boot Project

1. Go to https://start.spring.io/

2. Configure:
   - Project: Maven
   - Language: Java
   - Spring Boot: 3.2.2 (or latest 3.x)
   - Group: com.taskflow
   - Artifact: taskflow-api
   - Packaging: Jar
   - Java: 21

3. Add Dependencies:
   - Spring Web
   - Spring Data JPA
   - H2 Database
   - Lombok

4. Click "Generate" and download
5. Extract the ZIP file to your computer
6. Open the extracted folder in VS Code

Your Tasks

TASK 1: Create Task Entity and Repository

Create file: src/main/java/com/taskflow/entity/Task.java

Copy this code:

```
package com.taskflow.entity;

import jakarta.persistence.*;
import lombok.*;
import java.time.LocalDate;

@Entity
@Table(name = "tasks")
@Getter
@Setter
@NoArgsConstructor
@AllArgsConstructor
public class Task {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    @Column(nullable = false)
    private String title;

    private String description;

    private String priority;

    private LocalDate dueDate;

    @Column(nullable = false)
    private Boolean completed = false;
}
```

Create file: src/main/java/com/taskflow/repository/TaskRepository.java

Copy this code:

```
package com.taskflow.repository;

import com.taskflow.entity.Task;
import org.springframework.data.jpa.repository.JpaRepository;
import org.springframework.stereotype.Repository;

@Repository
public interface TaskRepository extends JpaRepository<Task,
Long> {
}
```

TASK 2: Create TaskService with TODO

Create file: src/main/java/com/taskflow/service/TaskService.java

Copy this code and COMPLETE THE TODO:

```
package com.taskflow.service;

import com.taskflow.entity.Task;
import com.taskflow.repository.TaskRepository;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;
import java.util.List;

@Service
public class TaskService {

    @Autowired
    private TaskRepository taskRepository;

    public List<Task> findAllTasks() {
        return taskRepository.findAll();
    }
```

```
    // TODO: Complete this method
    // Use taskRepository.findById(id)
    // If not found, throw new RuntimeException("Task not found
with id: " + id)
    public Task findTaskById(Long id) {
        // YOUR CODE HERE
        return null;
    }

    public Task createTask(Task task) {
        return taskRepository.save(task);
    }
}
```

HINT for the TODO:

```
return taskRepository.findById(id).orElseThrow(() -> new
RuntimeException("Task not found with id: " + id));
```

TASK 3: Create TaskController with TODO

Create file: src/main/java/com/taskflow/controller/TaskController.java

Copy this code and COMPLETE THE TODO:

```
package com.taskflow.controller;

import com.taskflow.entity.Task;
import com.taskflow.service.TaskService;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.*;
import java.util.List;

@RestController
@RequestMapping("/api/tasks")
public class TaskController {

    @Autowired
```

```
    private TaskService taskService;

    @GetMapping
    public ResponseEntity<List<Task>> getAllTasks() {
        List<Task> tasks = taskService.findAllTasks();
        return ResponseEntity.ok(tasks);
    }

    // TODO: Complete this method
    // Call taskService.findTaskById(id)
    // Return ResponseEntity.ok(task)
    @GetMapping("/{id}")
    public ResponseEntity<Task> getTaskById(@PathVariable Long
id) {
        // YOUR CODE HERE
        return null;
    }

    @PostMapping
    public ResponseEntity<Task> createTask(@RequestBody Task
task) {
        Task createdTask = taskService.createTask(task);
        return new ResponseEntity<>(createdTask,
HttpStatus.CREATED);
    }
}
```

HINT for the TODO:

```
Task task = taskService.findTaskById(id);
return ResponseEntity.ok(task);
```

TASK 4: Configure Database

Edit file: src/main/resources/application.properties

Add these lines:

```
# Database configuration
spring.datasource.url=jdbc:h2:mem:taskflowdb
spring.datasource.driverClassName=org.h2.Driver
```

```
spring.datasource.username=sa
spring.datasource.password=

# JPA configuration
spring.jpa.hibernate.ddl-auto=update
spring.jpa.show-sql=true
spring.jpa.database-platform=org.hibernate.dialect.H2Dialect

# H2 Console
spring.h2.console.enabled=true
```

Testing Your Work

Step 1: Run the Application

```
mvn spring-boot:run
```

Step 2: Create a Task

```
curl -X POST http://localhost:8080/api/tasks \
  -H "Content-Type: application/json" \
  -d '{
    "title": "Complete Module 3",
    "description": "Finish assignment",
    "priority": "HIGH",
    "dueDate": "2026-05-20",
    "completed": false
  }'
```

Step 3: Get Task by ID

```
curl http://localhost:8080/api/tasks/1
```

Expected response:

```
{
  "id": 1,
  "title": "Complete Module 3",
  "description": "Finish assignment",
  "priority": "HIGH",
  "dueDate": "2026-05-20",
```

```
  "completed": false
}
```

Submission

Step 1: Create GitHub Repository

1. Go to github.com
2. Click "New repository"
3. Name it: module3-taskflow-api (or any name)
4. Make it PUBLIC
5. Click "Create repository"

Step 2: Push Your Code

Open Terminal/Command Prompt in VS Code:
1. In VS Code, open Terminal (View → Terminal or Ctrl+`)

2. Make sure you're in your Spring Boot project folder:
The terminal should show the path to your taskflow-api folder
Example: C:\Users\YourName\taskflow-api

3. Run these Git commands one by one:

```
git init
```

(This initializes Git in your project folder)

```
git add .
```

(This adds all your files)

```
git commit -m "Complete Module 3 assignment"
```

(This commits your files)

```
git branch -M main
```

(This renames the branch to main)

```
git remote add origin
https://github.com/YOUR-USERNAME/YOUR-REPO-NAME.git
```

(Replace YOUR-USERNAME and YOUR-REPO-NAME with your actual values)

```
git push -u origin main
```

(This uploads your code to GitHub)

Step 3: Submit to NTU Learn

1. Copy your GitHub repository URL
2. Submit it in the assignment submission on NTU Learn

Done! 🎉